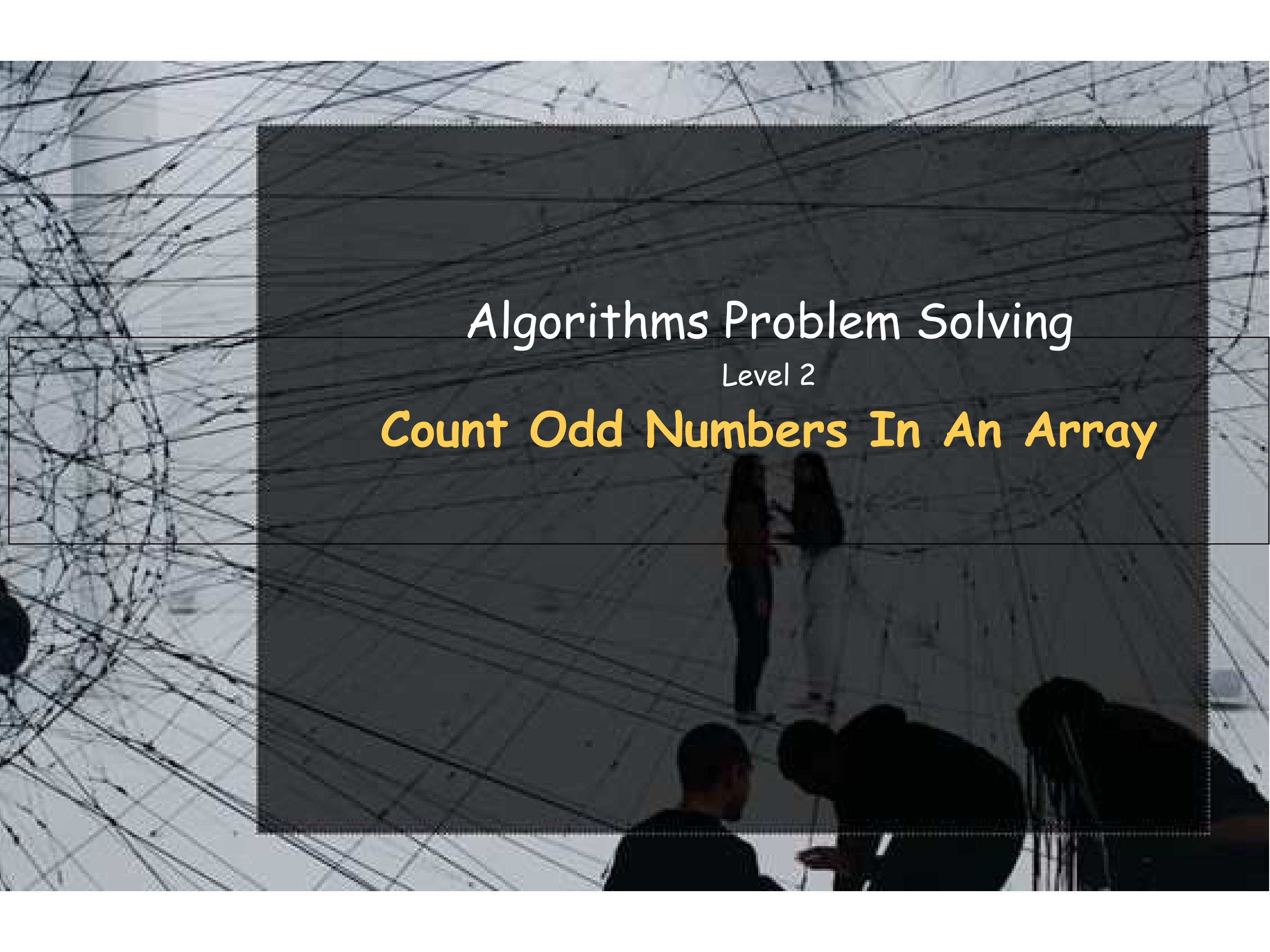


Algorithms Problem Solving Level 2

Count Odd Numbers In An Array



Algorithms Problem Solving

Level 2

Count Odd Numbers In An Array

Definition

- ◆ This program fills an array with random numbers from `1` to `100`.
- ◆ It checks each element to determine whether it is odd or even.
- ◆ It counts how many odd numbers exist in the array.
- ◆ `CountOddNumbers()` returns the total number of odd elements.

Explanation

- ◆ `CheckOddOrEven()` determines whether a number is odd or even.
- ◆ It uses the remainder operator `%`:
`Number % 2 != 0`
- ◆ If the remainder is not `0`, the number is odd.
- ◆ `CountOddNumbers()` starts with:
`int count = 0;`
- ◆ It then loops through all elements:
`for (int i = 0; i < arrLength; i++)`
- ◆ When an odd number is found, `count` is increased:
`count++;`
- ◆ After checking the whole array, the function returns `count`.

Steps in main()

1. Create an array with a maximum size of `100`.
2. Ask the user for the number of elements.
3. Fill the array with random numbers from `1` to `100`.
4. Print the array elements.
5. Set `count` to `0`.
6. Check each element using `CheckOddOrEven()`.
7. If the element is odd, increase `count`.
8. Continue until all elements have been checked.
9. Return the final count.
10. Print the number of odd elements.

💡 Example

◆ `Array Elements` :

42 68 35 1 70 25 79 59 63 65

◆ The odd numbers are:

35 1 25 79 59 63 65

◆ There are `7` odd numbers.

➡ Result

Array Elements : 42 68 35 1 70 25 79 59 63 65

Odd Numbers count is : 7

Note

- ◆ `CountOddNumbers()` does not create a new array.
- ◆ It only counts how many elements satisfy the condition.
- ◆ `count` is increased once for every odd number.
- ◆ The enum:

```
enum enOddOrEven { Odd = 1, Even = 2 };
```
- ◆ allows `CheckOddOrEven()` to clearly return whether a number is `Odd` or `Even`.
- ◆ `AddElementToArray()` is not used in this program.

⊘ Warning

- ◆ ``FillArrayWithRandomNumber()`` does not check that ``arrLength`` is ``100`` or less.
- ◆ Entering more than ``100`` elements can cause the program to access memory outside the array.
- ◆ ``ReadArrayElements()`` and ``AddElementToArray()`` are unused.
- ◆ ``cmath`` and ``corecrt_math_defines.h`` are unnecessary.
- ◆ ``srand()`` and ``time()`` should ideally have their proper headers included explicitly.
- ◆ The output says ``Array 1 Elements``, although there is only one array; ``Array Elements`` would be clearer.



Comparison

- ▣ `CheckOddOrEven()`
 - ✓ Checks one number.
 - ✓ Returns `Odd` or `Even`.
 - ✓ Does not count or modify the array.

- ▤ `CountOddNumbers()`
 - ✓ Checks all elements in the array.
 - ✓ Uses `CheckOddOrEven()` for each element.
 - ✓ Counts only the odd numbers.
 - ✓ Returns the final count.

✦✦ Summary

- ✓ Fill the array with random numbers.
- ✓ Check every element to determine whether it is odd.
- ✓ Increase `count` for every odd number.
- ✓ Return the final count.
- ✓ For the example array, there are `7` odd numbers.

Thank you 😊